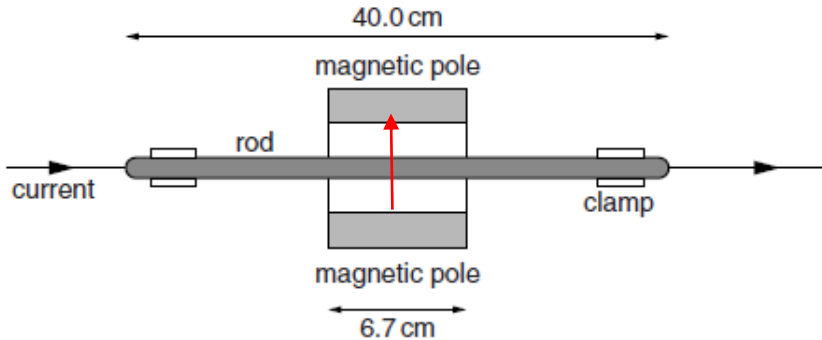


7	(a)	(i)	From definition, $V = \frac{W}{Q} \Rightarrow W = VQ$	1
		(ii)	From definition of work, $W = Fd$	1
	(b)	From $VQ = Fd$, we have Electric field strength (force per unit charge) = $\frac{F}{Q} = \frac{V}{d}$ (potential gradient since the potential changes linearly with distance between the plates)		1 1
	(c)	(i)	$(82.4 - 82.0) \times 10^{-3} \times 9.81 = 3.9 \times 10^{-3} \text{ N}$	1
		(ii)	- The current (flow) in the rod produces magnetic field around it which interacts with the permanent field of the U-shaped magnet. - Due to the interaction, a downward force acts on the magnet while at the same time as a result of Newton's third law, an upward force acts on the rod. - The rod is fixed but the magnet (or balance) is moveable and so this additional force is recorded. - a correct reference to Fleming's LHR and Newton's 3rd Law Alternative: When a current flows in the rod placed in a B-field, a magnetic force is experienced by the rod [1] By N3L, an equal and opposite force is exerted on the magnet [1]	1 1
		(iii)	1. 	

			The force on the U-shaped magnet is inwards (into the plane of the paper) and so the force on the rod must be outwards (out of the plane). Based on Fleming's Left Hand Rule, the magnetic field is pointing vertically upwards. • arrow from lower to upper magnetic pole or • lower pole marked as the North pole / upper pole labelled South	1
		(iii)	2. Uses $L = 6.7 \text{ cm}$ Recall and correctly substitute in $F = B\tilde{I}L$ $I = \frac{F}{BL} = \frac{(3.9 \times 10^{-3})}{(28.6 \times 10^{-3})(6.7 \times 10^{-2})}$ $= 2.03 \text{ A}$	1 1
			3. With the current reversed, the forces are attractive (the magnet is partially supported by the magnetic force due to the current in the rod). Thus, the reading is reduced, New reading $= (82.0 - 0.4) = 81.6 \text{ g}$	1

8	(a)		The steady direct voltage value which provides the same power / energy	1
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